Assignment 3: Hospital Billing

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Process healthcare bills for different types of patients. Sum up consultation, diagnostic, and treatment charges. Apply conditional concessions based on patient type (Inpatient/Outpatient) and total bill limits, classifying the final payments accordingly.

Algorithm

1. Store patient information in a nested list format containing Patient ID, Type, Consultation, Diagnostic, and Treatment costs.
2. Loop through each patient record.
3. Sum up the three bill components.
4. Use conditional branching: If Inpatient and bill exceeds 10,000, grant a 10% concession. If Outpatient and diagnostics exceed 2,000, waive 5% of diagnostics.
5. Determine final category string using if-statements based on the net bill.

Python Implementation

```
# Format: [PatientID, Type, Consultation, Diagnostics, Treatment]
patients = [
    ["PID-001", "Outpatient", 800, 2500, 0],
    ["PID-002", "Inpatient", 1500, 4000, 12000],
    ["PID-003", "Outpatient", 500, 1000, 500]
]

for p in patients:
    pid, p_type, consult, diag, treat = p
    gross_total = consult + diag + treat
    concession = 0.0

    # Conditional Concession Policy
    if p_type == "Inpatient":
        if gross_total > 10000:
            concession = gross_total * 0.10
    elif p_type == "Outpatient":
        if diag > 2000:
            concession = diag * 0.05

    net_bill = gross_total - concession

    # Classification Logic
    billing_tier = ""
    if net_bill > 15000:
        billing_tier = "Tier A (High Value)"
    elif net_bill > 5000:
        billing_tier = "Tier B (Medium Value)"
    else:
        billing_tier = "Tier C (Standard)"

    # Pattern Generation / String formatting
    print("-" * 40)
    print("Patient ID:", pid)
    print("Type      :", p_type)
```